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REPORT

Epidemiology of skeletal eroding band on the Great Barrier Reef and the role of injury in the initiation of this widespread coral disease

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Abstract Skeletal eroding band (SEB), which manifests as dense aggregations of the ciliate *Halofolliculina corallasia*, was the first coral disease described from the Indo-Pacific. Little is known about its etiology or impact. This study describes the distribution, prevalence and host range of SEB on a 500 km extent of the Great Barrier Reef (GBR), together with in situ rates of progression and infection following experimental injury. SEB occurred on 90–100% of reefs surveyed (n=18) in each of 3 years, demonstrating that SEB is widely distributed and persistent. SEB had the highest prevalence of any disease, affecting approximately 2% of 283,486 scleractinians and hydrocorals surveyed. Its host range was large, affecting 12 families and at least 82 scleractinian species, as well as the hydrocoral, *Millepora*. Corals in the families Pocilloporidae and Acroporidae were most susceptible, the former being up to five times more susceptible than other families. Progressive tissue loss was recorded on 95% of *Acropora muricata* colonies monitored ($n = 18$), with rates of SEB progression averaging ~2 mm/day. Injury experiments demonstrated that *H. corallasia*, the putative pathogen of SEB, readily colonised recently exposed coral skeleton in the absence of a vector, but did not colonise intact coral tissue. Invading ciliates failed to form band-like aggregations associated with progressive tissue loss on any of three coral species tested experimentally, suggesting that, while *H. corallasia* readily colonises recently exposed coral skeleton, it may not be sufficient in itself to cause tissue mortality. Interactions with additional agents or factors, increasing ciliate virulence or lowering disease resistance of coral hosts may be required before halofolliculinid infections become associated with tissue loss.

Keywords Skeletal eroding band (SEB) · *Halofolliculina corallasia* · Coral disease · Epizoism · Injury · Great Barrier Reef

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Introduction

Recognition of the serious impacts of coral disease on reef assemblages is increasing worldwide, commensurate with the increasing numbers of reports (Ward and Lafferty 2004) and diseases described in the last three decades (Green and Bruckner 2000; Raymundo et al. 2003; Sutherland et al. 2004; Willis et al. 2004; Weil et al. 2006). Until recently, research has focused primarily on coral diseases in the Caribbean, where they have caused significant and widespread mortality of both scleractinian (Williams et al. 1999; Porter et al. 2001; Patterson et al. 2002) and gorgonian corals (Guzman and Cortes 1984; Garzon-Ferreira and Zea 1992; Nagelkerken et al. 1997; Kim and Harvell 2004). Growing realisation that disease commonly plays a role in the ecology of many Indo-Pacific coral species (Loya 2004; Willis et al. 2004; Winkler et al. 2004; Raymundo et al. 2005; Dalton and Smith 2006; Page and Willis 2006) highlights the importance of understanding the etiology and impacts of disease, particularly given recent increases in the number of diseases (Raymundo et al. 2003; Willis et al. 2004) and disease cases in the region (Antonius and Lipscomb 2001; Willis et al. 2004).

Skeletal eroding band (SEB) was the first coral disease described from an Indo-Pacific reef (Antonius 1999), however, little is known beyond its macroscopic appearance and aspects of the biology of the putative pathogen. The disease manifests as a black band (~1–10 cm wide) at the interface between recently exposed skeleton and apparently healthy coral tissue (Fig. 1a–d; Antonius 1999; Antonius and Lipscomb 2001). The black band may appear speckled, depending on the density of the sessile folliculinid ciliate, *Halofolliculina corallasia*, which comprises the band. The ciliates embed themselves in the coral skeleton producing an eroded appearance. Koch's postulates have not been fulfilled for this disease, however, coral mortality is thought to be caused by spinning and chemical secretions of the asexually produced motile swarming phase of *H. corallasia* (Antonius and Lipscomb 2001). The band of ciliates is so dense at times that SEB may be confused with black band disease (BBD) in the field (Antonius and Lipscomb 2001). Recent apparent increases in the geographical distribution of SEB in the Indo-Pacific may be partly explained by the increasing familiarity of researchers with the distinction between SEB and BBD.

Since its discovery in 1988, SEB has been recorded from reefs throughout the Indo-Pacific, including Motupore Island (Papua New Guinea), Lizard Island (Great Barrier Reef [GBR]), Mauritius, and the Red Sea (Antonius and

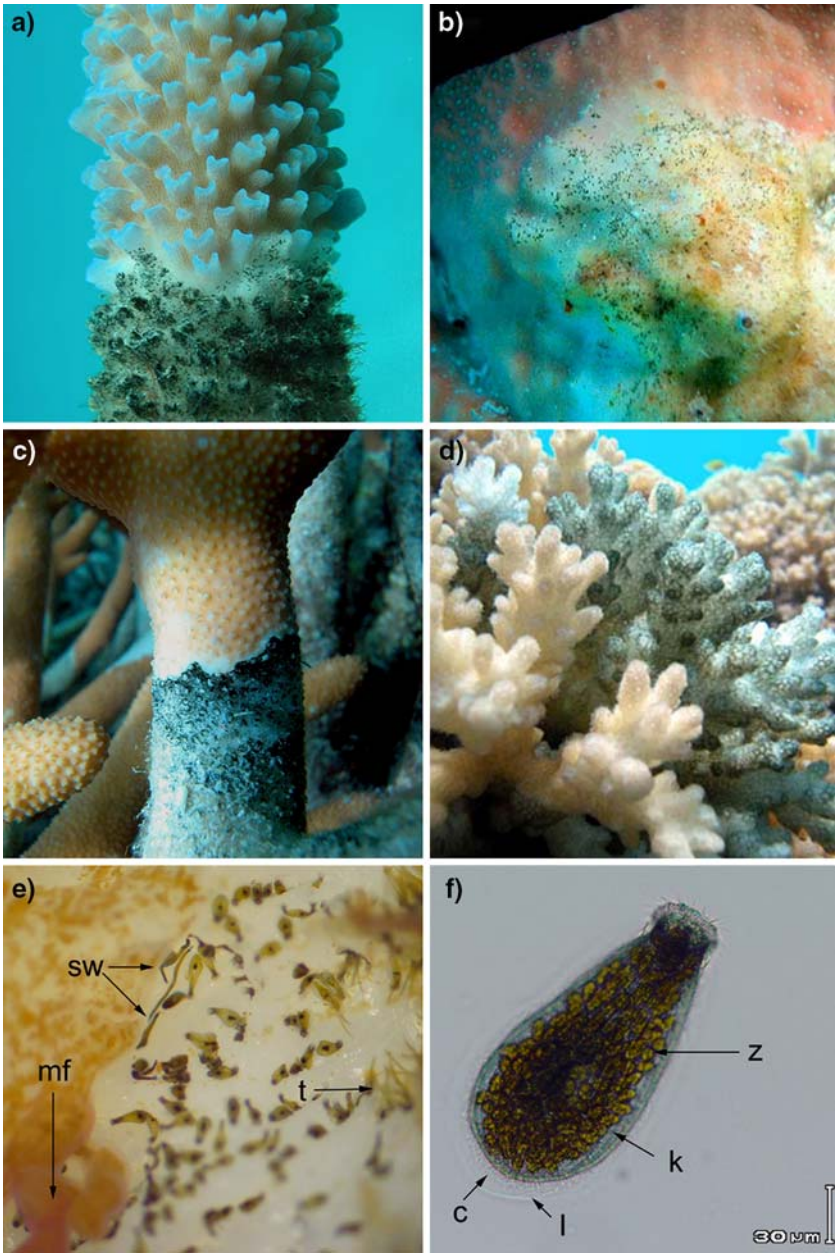

Fig. 1 Macroscopic field signs of skeletal eroding band (SEB) on **a** *Acropora pulchra*, **b** *Turbinaria stellata*, **c** *Acropora muricata*, and **d** *Pocillopora damicornis*. Microscopic images of **e** motile swarmers (sw) of *Halofolliculina corallasia* ahead of sessile trophonts (t) at the interface between live coral and bare skeleton on *P. damicornis*. [Note that the coral tissue is damaged and a mesenterial filament (mf) is exposed] and **f** motile swarmer undergoing morphogenesis and secreting a lorica (l). Cilia (c) which lie in rows or kineties (k) and zooxanthellae (z) are clearly visible

Lipscomb 2001). Its distribution continues to widen with recent records from the northern and southern GBR (Willis et al. 2004), New Britain (Papua New Guinea; Neale, personal observation), Palau, the Marshall Islands, the Solitary Islands (Willis and Page, personal observation) and the Philippines (Harvell et al. 2007). SEB has not been detected on Indo-Pacific reefs in Indonesia, Guam or Moorea, or on Caribbean reefs in Belize, the Florida Keys or the Dominican Republic, in surveys between 1997 and 2000 (Antonius and Lipscomb 2001) or more recently (Weil 2004). However, another ciliate of the genus *Halofolliculina* was recently found on Caribbean corals (Cróquer et al. 2006a), although it appears to be a different species (Cróquer et al. 2006b). In this study, therefore, the term SEB will not include Caribbean ciliate infections.

Compared to other coral diseases, the prevalence of SEB is extremely high. It was the most prevalent scleractinian disease on two of three GBR reefs surveyed in 2002–2003, affecting 5.4% of all corals (Willis et al. 2004). Moreover, it was found on up to 38% of corals in Red Sea surveys (Winkler et al. 2004). Despite the limited number of reefs in these surveys, SEB has been found on at least 31 scleractinian species on the GBR (Willis et al. 2004) and 22 genera in the Red Sea (Winkler et al. 2004). To date, only BBD and the white diseases (white plague-like diseases/white syndrome) have greater host ranges (summarised in Sutherland et al. 2004). Rates of partial and whole colony mortality caused by SEB are unknown. Progression rates of the band have been estimated to vary between 1 mm/week and 1 mm/day (Antonius and Lipscomb 2001), but quantitative measurements are lacking.

Observations of "loose settlements" of *H. corallasia* scattered across recently exposed coral skeleton suggest that the ciliate opportunistically settles following tissue loss caused, for example, by predation or physical injury (Winkler et al. 2004). This interpretation is supported by the positive correlation found between the prevalence of SEB and the prevalence of coral damage or injury on reefs in the Red Sea (Winkler et al. 2004). However, currently, it is unclear if loose settlements of ciliates lead to tissue loss and skeletal erosion associated with band-like aggregations of the ciliates. Furthermore, it is unclear if *H. corallasia* is the causative agent of tissue loss or if the ciliate colonises skeleton exposed by another micro-organism.

The goal of this study was to investigate the etiology, impact, and role of injury in the initiation of SEB on corals of the GBR. Specific objectives were to (1) determine the distribution, prevalence and host range of SEB from large-scale surveys of the GBR, (2) determine within colony rates of disease progression, (3) evaluate the role of injury in the initiation of SEB, and (4) determine if an intermediate host is required for transmission of *H. corallasia* to recently injured corals.

Materials and methods

SEB distribution, prevalence, host range and rates of disease progression

To determine the prevalence and host range of SEB on the GBR, surveys were completed on 18 reefs spanning 500 km in each of 3 years (2004, 2005, 2006). To ensure representation of reef types and habitats, reefs were surveyed within each of three cross-shelf positions (inner-, mid- and outer-shelf) in each of two latitudinal sectors (Cooktown /Lizard Island and Townsville sectors; Fig. 2). Reefs were surveyed during austral summers, when prevalence and host range were likely to be highest (Willis et al. 2004). Within each cross-shelf by sector position, two sites on each of three reefs were surveyed. Surveys involved three replicate 20 m × 2 m belt transects placed parallel to depth contours on the upper reef slope at 3–6 m depth (as per Willis et al. 2004; Page and Willis 2006). Reefs were selected haphazardly from those accessible within sectors. Sites were selected haphazardly from within areas characterised as northwestern, back reef slopes and transects were placed haphazardly on upper reef slopes. Within each 20 m × 2 m belt, all scleractinians, gorgonians, alcyonaceans and hydrocorals were examined, the presence or absence of SEB was recorded, and corals identified to the lowest taxonomic or morphological group, as appropriate. Differences in susceptibility to SEB among taxa were tested using a χ^2 test of independence. Other diseases on corals showing signs of SEB were also recorded.

Rates of partial mortality attributable to SEB were examined on colonies of *Acropora muricata* at Lizard Island (S 14°21′ 15.55, E 145°26′ 36.85), a mid-shelf reef located in the Cooktown/Lizard Island sector of the GBR. Eighteen branches, each from a separate colony, were tagged using numbered plastic cattle tags secured with cable ties to exposed skeleton. Tags were attached a short distance behind the disease front to avoid interfering with disease progression. The distance from cable ties to the nearest live tissue was measured in late November 2004 and again 4 weeks later in late December. The difference between the two measurements was used as a measure of linear disease progression and divided by the number of days between measurements to calculate a daily rate of SEB progression.

In situ injury experiments

A pilot study to test the most appropriate position for artificial injuries on coral branches and the appropriate timespan for following ciliate colonisation was completed on shallow patch reefs (2–5 m) at Lizard Island in November 2004. To determine whether intact coral tissue completely surrounding an injury is an impediment to ciliate colonisation and

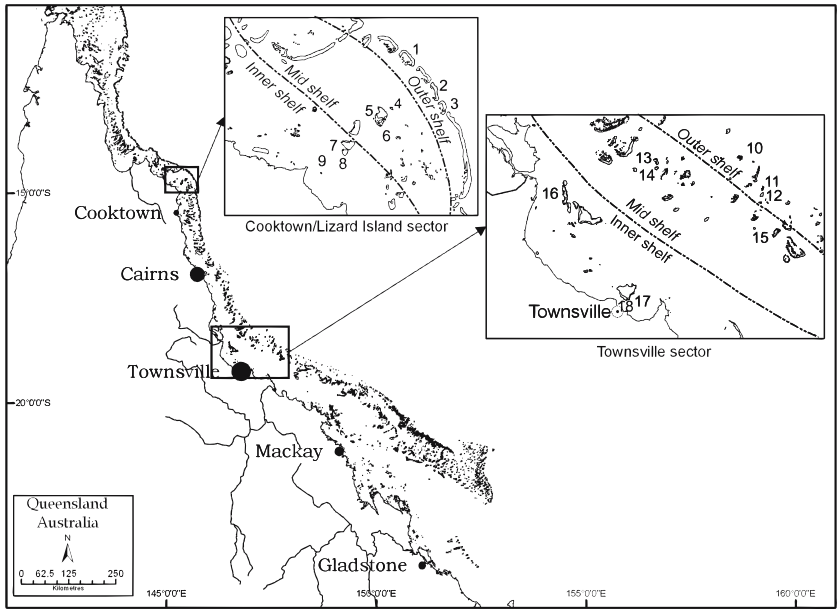

Fig. 2 Reefs surveyed annually between 2004 and 2006 for SEB in two latitudinal sectors of the Great Barrier Reef. Reefs are numbered as follows: *1* Day Reef, *2* Yonge Reef, *3* No Name Reef, *4* Macgillivray Reef, *5* Lizard Island, *6* North Direction Island, *7* Martin Reef, *8* Linnet Reef, *9* Maxwell Reef, *10* Dip Reef, *11* Knife Reef, *12* Fork Reef, *13* Kelso Reef, *14* Little Kelso Reef, *15* Davies Reef, *16* Orpheus Island, *17* Magnetic Island, *18* Middle Reef

hence the initiation of SEB, artificial injuries were created in two positions: one completely surrounded by intact coral tissue (mid-branch injury), and one immediately adjacent to exposed skeleton at the base of branches (basal injury). Ciliate densities were assessed 7, 14 and 44 days after injuries were inflicted.

Artificial injuries were created on branches in situ using an airgun attached to a SCUBA tank. High-pressure air removed coral tissue, but the underlying skeleton was not visibly damaged, mimicking the removal of coral tissue by corallivorous starfish (*Acanthaster planci)* or gastropods (*Drupella* spp.). The size of the injury was restricted using a square plastic shield with a 2 cm × 2 cm window held in front of the coral branch. Injuries in the two branch positions (basally and mid-branch) were inflicted on 20 physiologically separate branches of three scleractinian species commonly affected by SEB: *A. muricata, Pocillopora damicornis* and *Porites cylindrica*. Artificially injured branches and an equal number of uninjured control branches (20 per species) were tagged using plastic cable ties to assist in their relocation. Five injured and five uninjured branches of each species were then collected 7, 14 and 44 days after injury using bone cutters. Each branch was placed in a separate sealed plastic bag and transported immersed in seawater. All injured and non-injured branches were examined microscopically, and the number of ciliates per injury (injured branches) or per 2 cm × 2 cm area (uninjured branches) was recorded.

Differences in ciliate densities among coral species and between injury positions over time (7 vs. 14 day samples) were examined using a three-factor, fully orthogonal ANOVA, with coral species, injury position and time treated as fixed factors. In addition, a two-factor ANOVA was used to examine differences in ciliate densities on mid-branch injuries among species over the three sampling periods (7, 14 and 44 days after injury). Basal injuries were excluded from the second analysis because they were generally obscured by heavy fouling 14 days after injury. For this reason, only mid-branch injuries were used in subsequent injury experiments.

A full-scale experiment was conducted at Lizard Island over the austral summer of 2004/2005 to determine rates of colonisation of artificial injuries by ciliates, changes in ciliate densities through time, levels of coral mortality associated with their presence, and if the band-like manifestation of SEB developed following ciliate colonisation of injuries. Artificial injuries were inflicted mid-branch, as described above, on 90 branches of each of *A. muricata, P. damicornis* and *P. cylindrica*. 90 uninjured branches of each species were also tagged (experimental controls). Injured and uninjured branches were examined in situ on days 1, 5, 10, 15, 20 and 133 and all injuries remeasured (maximum and perpendicular diameters). At each sampling, 15 randomly selected injured and uninjured branches of each species were collected and examined microscopically, as described above. Numbers of ciliates on the entire injury were counted when densities were low or in six replicate 5 mm × 5 mm areas subsampled haphazardly within each injury site when densities were high. Ciliate densities were then standardised to the number per cm^2.

The effects of coral species and time on the proportion of injuries with ciliates present were tested using the linear model: Proportion with ciliates = Species + Time + Species × Time. The response variable was the binary classification of injuries with ciliates either present or absent. Data were analysed using logistic regression with categorical

variables, where the logit (the log of the ratio of probability of ciliates present vs. ciliates absent) was the link function (McCullagh and Nelder 1989). A two-way fixed factor ANCOVA controlling for injury size was used to test for changes in ciliate densities through time and between species. Single degrees of freedom ANCOVA and Tukey's HSD tests were subsequently used to identify when changes in ciliate densities occurred in each species. Changes in injury size on the 15 branches of each species collected 133 days after injury were examined using a multivariate repeated measures ANOVA, followed by single degrees of freedom ANOVA and Tukey's HSD tests to determine the timing of significant changes in injury size.

Aquarium injury experiment

To evaluate the possibility that other reef organisms, particularly fish and corallivores such as *Drupella* spp., act as vectors and transmit ciliates to injuries, an exclusion experiment was carried out in aquaria. Thirty branches of *A. muricata, P. damicornis* and *P. cylindrica* were collected from the field experimental site and transported immersed in seawater to the laboratory. Branches were cleaned of all visible microfauna and flora. Each broken branch edge was covered with non-toxic modelling clay and placed in a 3 cm long section of PVC piping, which held branches upright in racks. Ten branches of each species were randomly assigned to one of three aquaria and mid-branch injuries (2 cm × 2 cm) were created on five randomly chosen branches of each species in each aquarium. Thus, each aquarium contained five injured and five uninjured branches of each coral species. Aquaria were supplied with a high flow of unfiltered seawater pumped from the local reef flat and fragments were examined microscopically for the presence of ciliates 5 days after injury.

Results

Distribution, prevalence, host range and rate of disease progression

SEB was present on all reefs surveyed ($n = 18$) in 2005 and on all but one reef (Middle Reef in the Townsville sector) in 2004 and 2006. SEB accounted for the majority (40–60%) of disease cases recorded in each year (data for other diseases not shown), making SEB the most prevalent coral disease in regions surveyed. SEB predominantly affected scleractinian corals, but also at least one species of *Millepora* (O: Hydrocorallina) (Table 1); it was not observed to affect alcyonaceans, gorgonians or antipatharians. SEB affected between 1.2% (1,101 of 91,552 colonies in 2006) and 2.3% (2,177 of 96,148 colonies in 2005) of combined scleractinian and hydrocoral taxa surveyed each year. At the transect level, SEB prevalence ranged from 0 to 11.9% of combined scleractinian and hydrocoral taxa.

SEB affected a wide range of coral hosts, including 12 families, 26 genera and 81 species of scleractinians, plus at least one species of *Millepora* (O: Hydrocorallina) (Table 1). Susceptibility to SEB varied significantly among scleractinian and hydrocoral families ($\chi^2 = 2780.93$, $df = 12$, $p < 0.001$). Pocilloporid corals were ~1.5 times more susceptible than acroporids, ~5 times more susceptible than poritids and greater than five times more susceptible than corals in all other families (Table 1, Fig. 3). Although pocilloporids were the most vulnerable to SEB, 57% of cases were recorded from acroporid corals, reflecting the greater abundance of this family in surveys. In comparison, 18% of cases were recorded from pocilloporid corals, 20% from poritid corals, 3.6% from faviids and less than 1% of cases from the remaining susceptible families. Colonies with signs of SEB also showed signs of other diseases, including tumors and cyanobacterial infections, however the prevalence of colonies with multiple disease signs was very low (Table 1).

In situ monitoring over 4 weeks revealed that SEB disease fronts progressed on all but one tagged branch of *A. muricata* ($n = 18$ branches). Rates of linear progression ranged from 0.3 to 3.3 mm/day, with an average (±SE) progression rate of 2 ± 0.3 mm/day ($n = 17$ branches) in December 2004.

In situ injury experiments

Halofolliculina corallasia colonised injuries created on all three experimental coral species in the pilot field study (Table 2), whereas no ciliates were observed on control branches or on healthy parts of injured branches, suggesting that injury enhances the ability of ciliates to colonise corals. The suitability of exposed skeleton for ciliate colonisation differed significantly among coral species, injury position and time after injury, with *A. muricata* and *P. cylindrica* being preferred hosts. A total of 100% of injuries on *A. muricata* and *P. cylindrica* were colonised within 7 days, whereas only 70% of injuries on *P. damicornis* were colonised by this time. Ciliates were recorded on both basal and mid-branch injuries, the latter indicating that ciliates either traverse live coral tissue to reach injured areas or settle directly onto exposed skeleton from the water column. Patterns in ciliate densities for the two injury positions differed among the three species ($\text{ANOVA}_{(\text{Injury position} \times \text{Species})}$: $p < 0.001$, Table 3). On *A. muricata*, basal injuries were colonised by higher densities of ciliates than mid-branch injuries (Tukey's HSD: $p < 0.001$), while on *P. damicornis* and *P. cylindrica*, ciliate densities did not differ significantly between injuries at the two branch positions.

Table 1 Prevalence and host range of skeletal eroding band (SEB) in annual surveys of 18 reefs of the Great Barrier Reef between 2004 and 2006

Family	Species or taxonomic group	Total # of colonies with SEB	Total # of colonies with SEB + other diseases	Total # of colonies surveyed	Prevalence of SEB (%)	Prevalence of SEB + other disease (%)	Records of species or taxa with SEB
Pocilloporidae		954	47ab	20,316	4.7	0.002	X * ^
	Other Pocilloporids	85	2ab	997	8.5	0.002	X * ^
	Pocillopora eydouxi						X
	Pocillopora meandrina						X
	Pocillopora verrucosa						X * ^
	Seriatopora spp.	259	15a	4,445	5.8	0.003	X *
	Seriatopora hystrix		a				X *
	Pocillopora damicornis	181	5a	4,213	4.3	0.001	X * ^
	Stylophora pistillata	429	25a	10,661	4	0.002	X * ^
Acroporidae		3,067	32ab	100,234	3.1	<0.001	X * ^
	Bottlebrush *Acropora*	559	2a	10,090	5.5	<0.001	X *
	Acropora echinata						X
	Acropora longicyathus		a				X
	Staghorn *Acropora*	1,776	15a	40,898	4.3	<0.001	X * ^
	Acropora florida						X ^
	Acropora grandis						X *
	Acropora intermedia		a				X *
	Acropora micropthalma						X *
	Acropora muricata		a				X * ^
	Acropora pulchra						X
	Acropora robusta						X
	Acropora yongei						X *
	Tabulate *Acropora*	132	3ab	3,725	3.5	0.001	X * ^
	Acropora clathrata						X ^
	Acropora cytherea						X *
	Acropora downingi						^
	Acropora hyacinthus		a				X * ^
	Clumping *Acropora*	20	4a	534	3.7	0.007	X
	Acropora listeri						X
	Bushy *Acropora*	78	1a	2,531	3.1	<0.001	X * ^
	Acropora aspera						^
	Acropora austera						X *
	Acropora elseyi						X

Table 1 continued

Family	Species or taxonomic group	Total # of colonies with SEB	Total # of colonies with SEB + other diseases	Total # of colonies surveyed	Prevalence of SEB (%)	Prevalence of SEB + other disease (%)	Records of species or taxa with SEB
	Acropora horrida						X
	Acropora loripes						X *
	Acropora sarmentosa						X
	Acropora valida						X * ^
	Corymbose *Acropora*	223	5a	9,280	2.4	0.001	X *
	Acropora aculeus						X
	Acropora chesterfieldensis						X
	Acropora divaricata						X
	Acropora donei						X
	Acropora kimbiensis		a				X
	Acropora latistella		a				X
	Acropora lutkeni						X
	Acropora millepora		a				X *
	Acropora nasuta		a				X *
	Acropora secale						X
	Acropora spathulata		a				X
	Acropora subulata						X
	Acropora tenuis						X ^
	Digitate *Acropora*	122		7,246	1.7		X *
	Acropora digitifera						X
	Acropora gemmifera						X *
	Acropora humulis						X *
	Acropora monticulosa						X *
	Isoporan	100		10,303	1		X *
	Acropora bruggemanni						X
	Acropora palifera						X *
	Astreopora	1		1,349	0.1		X *
	Montipora	56	2a	14,186	0.4	<0.001	X * ^
	Montipora hispida						X
	Montipora monasteriata						^
	Montipora spumosa		a				X

Table 1 continued

Family	Species or taxonomic group	Total # of colonies with SEB	Total # of colonies with SEB + other diseases	Total # of colonies surveyed	Prevalence of SEB (%)	Prevalence of SEB + other disease (%)	Records of species or taxa with SEB
Poritidae		1,074	10ab	102,892	1	<0.001	X *
	Branching *Porites*	897	12ab	60,579	1.5	<0.001	X
	Porites cylindrica		a				X
	Porites nigrescens						X
	Massive *Porites*	123	a	15,166	0.8		X *
	Other *Porites*	54		15,805	0.3		X
	Porites lichen						X
	Porites rus						X
	Goniopora/Alveopora	0		11,342			X *
Merulinidae		14		1,865	0.8		X * ^
	Hydnophora exesa						X
	Hydnophora microconos						^
	Hydnophora rigida						X *
Faviidae		196	1a	36,537	0.5	<0.001	X * ^
	Other Faviidae	149	1a	13,514	1.1	<0.001	X *
	Cyphastrea chalcidium						^
	Cyphastrea serailia						X ^
	Echinopora gemmacea						X
	Echinopora horrida			a			X *
	Echinopora lamellose						X *
	Echinopora mammiformis						X
	Diploastrea heliopora						X
	Leptastrea purpurea						^
	Leptoria phrygia						X
	Platygyra/Goniastrea	38		10,872	0.4		X *
	Platygyra daedalea						X
	Platygyra meandrina						X
	Goniastrea aspera						X
	Goniastrea edwardsi						X
	Goniastrea retiformis						X ^
	Favia/Favites/Montastrea	9		12,151	0.1		X *
	Favia pallida						X
	Favia stelligera						X ^
	Favites abdida						^

Table 1 continued

Family	Species or taxonomic group	Total # of colonies with SEB	Total # of colonies with SEB + other diseases	Total # of colonies surveyed	Prevalence of SEB (%)	Prevalence of SEB + other disease (%)	Records of species or taxa with SEB
Oculiniidae		29		5,214	0.6		X *
	Galaxea astreata						X
	Galaxea fascicularis						X
Pectiniidae		4		829	0.5		X *
	Mycedium elephantotus						X
Mussidae		9	1a	2,866	0.3	<0.001	X *
	Lobophyllia corymbosa						X
	Lobophyllia hemprichii						X
	Scolymia sp.						X
	Symphyllia sp.						X
Dendrophyllidae		3		1,124	0.3		X
	Turbinaria retiformis						X
	Turbinaria stellulata						X
Siderastreidae	*Coscinarea*	1		1,065	0.1		X
Agariciidae		1		1,899	0.1		X * ^
	Pachyseris rugosa						^
Fungiidae	*Podabacia crustacea*	1		4,217	0.02		X *
Milleporidae (O:Hydocorallina)		16		4,068	0.4		X *
	Millepora sp.						X
	Millepora tenella						X
	Number of families affected	13					
	Minimum # of species affected	82					

Taxa showing signs of SEB, either alone or in combination with other disease signs, are listed. X denotes records of SEB from this study; * denotes records of SEB from Willis et al. (2004); and ^ denotes records of SEB from Antonius and Lipscomb (2001). Online lowercase letters denote other diseases recorded from colonies with signs of SEB: a = other cyanobacterial infections, b = tumours

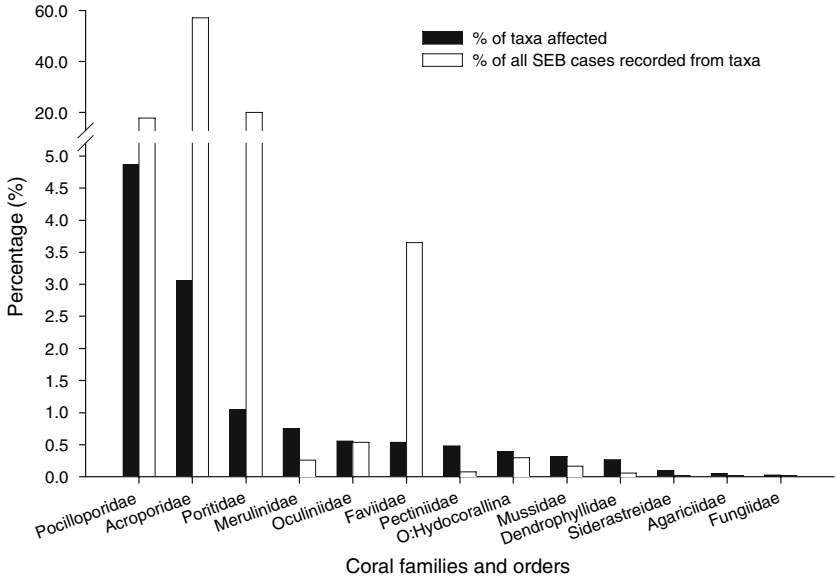

Fig. 3 Prevalence of SEB within each family and order (*black bars*) and the percentage of total SEB cases recorded from each family and order (*white bars*). Families and Order shown in rank order of SEB prevalence

Table 2 Colonisation of mid and basal branch injuries created on three coral species (*Acropora muricata, Porites cylindrica* and *Pocillopora damicornis*) by the ciliate *Halofolliculina corallasia*

Species	Parameter (ciliate density per cm^2, or % of injuries with ciliates)	Days after injury				
		7		14		44
		Mid-branch	Basal	Mid-branch	Basal	Mid-branch
A. muricata	Mean # ciliates	110.02 ± 95.8	425.07 ± 120.8	386.4 ± 55	661.33 ± 238.1	20.55 ± 13
	% of injuries with ciliates	100 (5)	100 (5)	100 (5)	100 (5)	3
P. cylindrica	Mean # ciliates	19.5 ± 6.6	14.1 ± 34.1	192.9 ± 47.5	340 ± 81.4	30.45 ± 1.6
	% of injuries with ciliates	100 (5)	100 (5)	100 (5)	100 (5)	5
P. damicornis	Mean # ciliates	6 ± 1.1	0.95 ± 5.8	2.25 ± 0.1	5.5 ± 5.5	0
	% of injuries with ciliates	100 (5)	40 (2)	80 (4)	20 (1)	0

Mean ciliate densities (± SE) per wound and % of injuries with ciliates are compared through time (7–44 days after injury) for each coral. Number of injuries with ciliates present shown in parentheses

Patterns in ciliate densities differed among the three coral species through time, both when densities were compared between the first two sampling times (7 and 14 days after injury) for the two injury positions (ANOVA$_{(\text{Time} \times \text{Species})}$: $p < 0.001$, Table 3) and when densities on mid-branch injuries alone were compared among all three sampling times (7, 14 and 44 days after injury) (ANOVA$_{(\text{Time} \times \text{Species})}$: $F = 6.213$, $df = 2$, $p < 0.001$). Given the problem of defining the area of basal injuries by day 44 because of heavy fouling, only mid-branch injuries were used in subsequent experiments.

The full-scale, field injury experiment confirmed that recently exposed skeleton is a necessary precursor for colonisation of otherwise healthy corals by *H. corallasia.* More than 90% of artificially-created lesions ($n = 45$) contained ciliates within 5 days (Fig. 4), whereas no ciliates were observed on control branches or on healthy areas of injured branches (not shown). Rates of injury colonisation again varied among the three coral species and through time (Fig. 4, Table 4). For *P. cylindrica*, 100% of injuries were colonised within 24 h and ciliates were recorded on all wounds except in the final 4.5-month sample, at which time

Table 3 Three factor ANOVA comparing densities of the ciliate *Halofolliculina corallasia* on mid and basal branch injuries created on three coral species (*Acropora muricata, Porites cylindrica* and *Pocillopora damicornis*), 7 and 14 days after injury

Source of variation	*df*	*F*	*p*
Time	1	4.356	0.042*
Injury position	1	1.277	0.264
Species	2	80.012	<0.001*
Time × injury position	1	0.048	0.828
Time × species	2	10.015	<0.001*
Injury position × species	2	10.031	<0.001*
Time × injury position × species	2	1.370	0.264

Data are log + 0.0001 transformed

* denotes significance at $\alpha = 0.05$

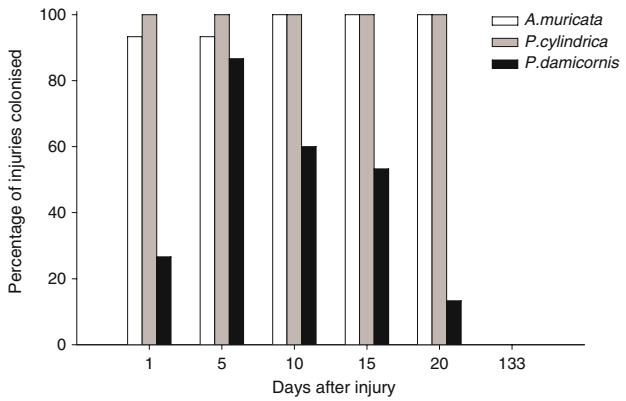

Fig. 4 Comparison of the percentage of artificially-created injuries having the ciliate, *Halofolliculina corallasia*, present among three coral species (*Acropora muricata, Porites cylindrica* and *Pocillopora damicornis*), 1–133 days after injury

Table 4 Logistic regression comparing the proportion of mid branch injuries colonized by the ciliate *Halofolliculina corallasia*, 1–133 days after injury

Factor	Model 1			Model 2 (saturated model)		
	df	χ^2	*p*	*df*	χ^2	*p*
Time	1	47.9655	<0.0001	2	12.8586	<0.01*
Species	2	47.8443	<0.0001	1	0.3940	0.53
Time × Species				2	7.4325	0.02*

Δ in Schwarz's Baysian criterion between model 1 and 2 = 109.32. Positive change in Schwarz's Baysian criterion indicates improvement of model by interaction term

* denotes significance at $\alpha = 0.05$

no ciliates were detected. For *A. muricata*, colonisation was slower, taking 10 days for 100% of injuries to be colonised, although again ciliates were present on all injuries prior to the final, 4.5-month sample. In contrast, although 86% of *P. damicornis* injuries were colonised by day 5, this proportion declined rapidly to ~13% within 3 weeks of injury, suggesting that exposed skeleton of *P. damicornis* represents a less suitable substrate for ciliate colonisation and population growth.

Patterns in the magnitude and timing of changes in mean ciliate densities varied among the three coral species ($\text{ANCOVA}_{(\text{Time} \times \text{Species})}$: $F = 22.694$, $df = 10$, $p < 0.001$). Similarly, overall mean ciliate densities differed among species ($\text{ANCOVA}_{(\text{Species})}$: $F = 177.5$, $df = 2$, $p < 0.0001$), further indicating that skeletons of the three species differ in their suitability for ciliate colonisation and replication (Fig. 5). Average ciliate densities were greatest on *P. cylindrica* injuries, reaching on average 834 ± 59 (±SE) ciliates per cm^2 20 days after injury, and were lowest on *P. damicornis* lesions, reaching on average only 41.9 ± 15.7 (± SE) ciliates per cm^2 by day 15. Although a more suitable host than *P. damicornis*, *A. muricata* was significantly less suitable than *P. cylindrica*, as indicated by the at least ten-fold fewer ciliates colonising *A. muricata* injuries (63 ± 10 [±SE] ciliates per cm^2). *P. cylindrica* clearly provided the most suitable substratum for ciliates, with mean ciliate densities at least doubling every 5 days until peaking at 20 days after injury (Figure 5). However, ciliate densities declined to zero by 4.5 months after injury on all three coral species. Ciliate declines on *P. damicornis* were commensurate with the healing of wounds in this species (Fig. 6i). The absence of ciliates on *A. muricata* and *P. cylindrica* injuries by the end of the study could be partly attributed to reductions of exposed skeleton through injury repair, however heavy fouling of the remaining exposed skeleton, predominantly by turfing and coralline algae (Fig. 6g,h), was primarily responsible for reduced densities of ciliates. Thus, although *H. corallasia* is a good early coloniser, it is a poor competitor relative to the general fouling community.

The presence of ciliates on coral skeleton exposed via injury did not lead to additional tissue loss on any of the injured branches. In fact, overall, the size of injuries declined through time (Fig. 7; $\text{ANOVA}_{(\text{Time})}$: Pillai's trace = 0.918, $F = 69.41$, $df = 6$, $p < 0.001$) indicating repair rather than degeneration of wounds in the presence of *H. corallasia* ciliates. The extent to which injuries healed varied among the three species, as indicted by variation in the timing and magnitude of declines in injury size among species (Fig. 7; $\text{ANOVA}_{(\text{Time} \times \text{Species})}$: Pillai's trace = 1.305, $F = 11.9$, $df = 12$, $p < 0.001$). All injuries on *P. damicornis* healed completely by 4.5 months, although tissue in the repaired area was visibly paler than surrounding tissues. In contrast, none of the lesions completely healed in the other two species. For *A. muricata*, there was no significant decrease in mean injury size over the

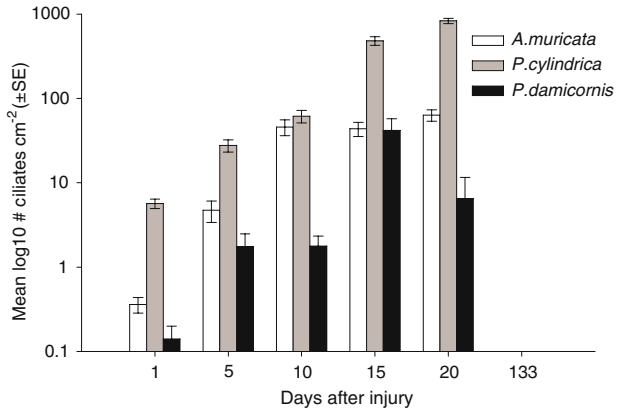

Fig. 5 Mean densities of the ciliate, *Halofolliculina corallasia*, on injuries artificially created on three scleractinian species (*Acropora muricata, Porites cylindrica* and *Pocillopora damicornis*), 1–133 days after injury

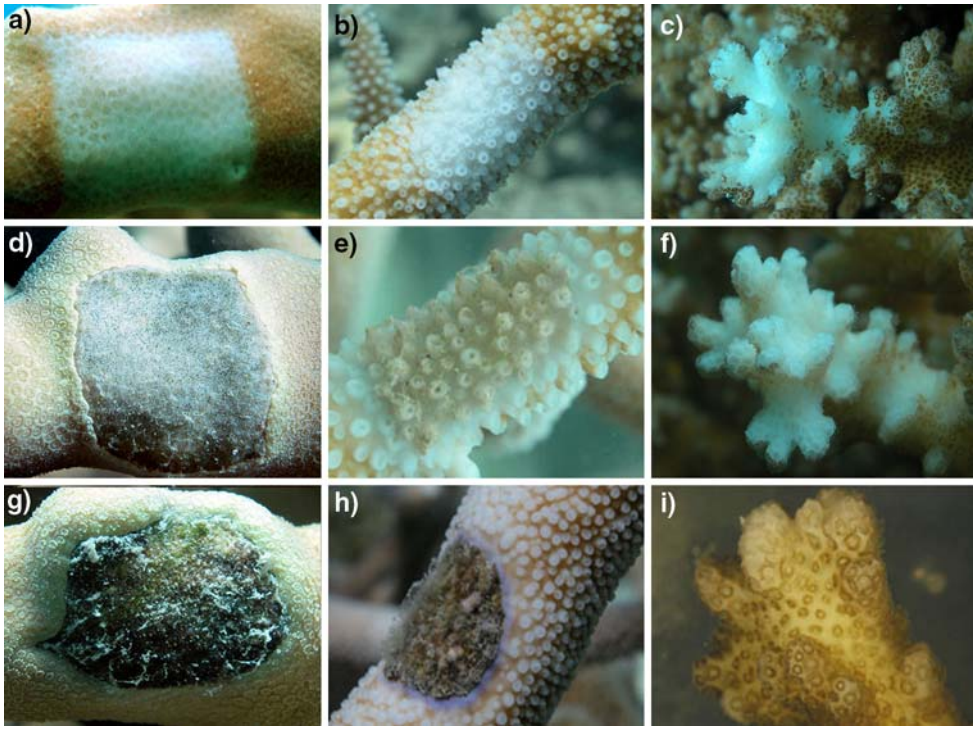

Fig. 6 Representative images showing temporal sequences in the colonisation of injuries artificially created on branches of (**a**, **d**, **g**) *Porites cylindrica*, (**b**, **e**, **h**) *Acropora muricata*, and (**c**, **f**, **i**) *Pocillopora damicornis* **a**–**c** bare white skeleton at time 0 following injury; **d**–**e** light fouling with some healing on all species at 10–20 days after injury [note high ciliate densities on *P. cylindrica* in **d**]; **g**, **h** healing along the edges of injuries, decreased injury size and heavy fouling of skeleton in *P. cylindrica* and *A. muricata* 133 days after injury; and **i** complete healing of injury in *P. damicornis* 133 days after injury

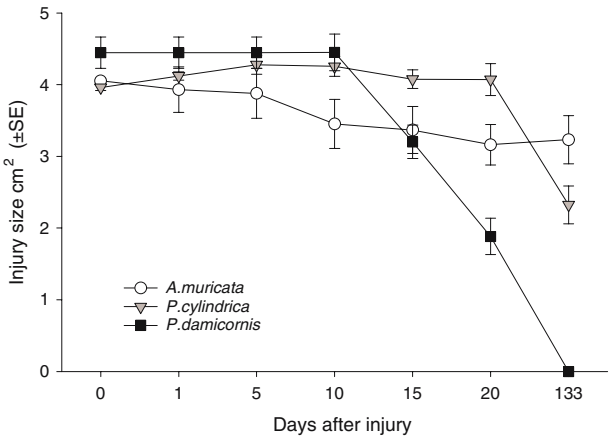

Fig. 7 Mean size of injuries, artificially created on three scleractinian species (*Acropora muricata, Porites cylindrica* and *Pocillopora damicornis*), 1–133 days after injury

4.5 months. Regeneration of injuries in *P. cylindrica* was relatively slow compared with regeneration in *P. damicornis*, with injury size declining by 40% between day 20 (4 ± 0.04 [±SE] cm^2) and 4.5 months after injury (2.3 ± 0.26 [±SE] cm^2).

Aquarium injury experiment

Exclusion of macroscopic organisms in the aquarium injury experiment confirmed that macroscopic vectors are not required for colonisation of newly exposed coral skeleton by *H. corallasia*. 85% of injured branches were colonised by ciliates in the absence of other organisms. Ciliates were again absent from areas of healthy tissue on both the injured and control branches.

Discussion

SEB distribution, prevalence, host range and rate of disease progression

Tissue loss associated with the progression of band-like aggregations of *H. corallasia* along branches of the staghorn coral, *A. muricata*, is recorded here and confirms that SEB infections are associated with progressive tissue loss of corals on the Great Barrier Reef (GBR). Mean and maximum rates of disease progression, of 2 ± 0.3 (± SE) and 3.3 mm/day, respectively, are low compared with rates of tissue loss caused by other coral diseases on the GBR. For example, mean rates of SEB progression are less than half those reported for BBD at Lizard Island (4.1–9.9 mm/day, Boyett et al. 2007) and are likely to be slower than those reported for white syndrome in the southern GBR (up to 57.8 mm^2/day, Roff et al. 2006), although faster than those recorded much further south (> 1,000 km south of the GBR) in the Solitary Islands (maximum of 0.52 mm/day, Dalton and Smith 2006). Although rates of disease progression are generally low for SEB, they are, nevertheless, within the range of progression rates recorded for several diseases that have caused significant mortality on Caribbean reefs (cf. 3–10 mm/day for BBD; 3–30 mm/day for

white plague II; 3.1 mm/day for white plague I (summarised in Weil 2004); and 6 mm/month for yellow band disease in the Caribbean, Cervino et al. 2001). Although rates of tissue loss caused by SEB may be at the lower end of the range for coral diseases, its widespread distribution suggests that it could nevertheless have a significant impact on coral assemblages.

The ubiquity of SEB on all reefs surveyed along more than 1,200 km of the GBR (combined data from this study and Willis et al. 2004) and on all reefs surveyed in the Red Sea (Winkler et al. 2004) suggests that SEB is endemic within these regions. The absence of SEB from one reef (Middle Reef, central GBR) in two survey years (2004 and 2006) is more likely to reflect its low abundance on this reef, rather than its complete absence in some years. The potential to miss a disease when its prevalence is low highlights the need for temporally and spatially replicated surveys to gain accurate distributional ranges of coral diseases. The widespread distribution of SEB, not only along the GBR, but also throughout the Indo-Pacific (Motupore: Antonius 1999 and New Britain, PNG: Neale, personal communication; Mauritius, Red Sea: Antonius and Lipscomb 2001; Philippines: Harvell et al. 2007; Palau, Marshall Is., Solitary Is.: personal observation), in combination with the recent discovery of a similar disease in the Caribbean (Cróquer et al. 2006b) suggest that halofolliculinid infections may be endemic in coral populations worldwide. As researchers become more familiar with field signs of SEB, its distribution range is likely to increase, particularly in the Indo-Pacific where coral disease has been understudied. SEB may be the only coral disease detectable in the fossil record due to the characteristic erosion of coral skeleton during lorica section by motile swarmers of *H. corallasia* (Riegl and Antonius 2003). Hence, the fossil record may provide clues on the emergence and spread of SEB on geological timescales.

The status of SEB as one of the most common coral diseases on the GBR has remained unchanged over the past 5 years (Willis et al. 2004; this study). In 2002–03, SEB accounted for 33% of disease cases recorded ($n = 1{,}800$) and was the most prevalent disease on two of the three reefs surveyed (Willis et al. 2004). Annual surveys of 18 reefs between 2004 and 2006 indicate that SEB is still the most abundant coral disease on the GBR, accounting for 40–66% of all disease cases. Although the overall prevalence of SEB is low, affecting only 2% of GBR corals surveyed, studies in the Red Sea demonstrate that it is capable of infecting up to 51% of corals on a reef, and up to 70% of pocilloporid corals in the genus *Seriatopora* (Winkler et al. 2004). In the present study, pocilloporid corals were also the most susceptible, with more than 4.5% of pocilloporids showing signs of SEB. Although acroporid corals were only three-quarters as likely to be infected on the GBR (~3.0% of acroporids showed signs of SEB), their greater abundance meant that they accounted for the majority (57%) of SEB cases. The greater susceptibility of corals in the family Pocilloporidae (1.5 times more susceptible than acroporids, 5 times more than poritids, and greater than 5 times more susceptible than corals in other families) meant that when combined with the Acroporidae, these two families accounted for ~75% of SEB cases. Interestingly, acroporid corals are the most common hosts for a number of other diseases, including black band (Page and Willis 2006) and brown band on the GBR (Willis et al. 2004) and white band in the Caribbean (Precht et al. 2002), highlighting the vulnerability of this family to disease.

Disease records from this study significantly expand the known host range for SEB. The 82 species with signs of SEB recorded is over 2.5 times greater than host ranges reported in previous smaller studies (Antonius and Lipscomb 2001; Willis et al. 2004). This increase in host range is undoubtedly a function of repeatedly surveying a wider range of reef types and coral communities than in previous studies, highlighting the important contributions that large-scale spatial and temporal studies make to epidemiological studies of coral disease. Continuation of large-scale surveys is required, however, to confirm that increased susceptibility of corals to SEB has not contributed to the documented increase in host range. SEB currently has a greater documented host range than any other coral disease worldwide (cf. Sutherland et al. 2004). As predicted above for its distributional range, its host range is likely to continue to increase as researchers worldwide become more familiar with its appearance in the field and as more reefs are surveyed.

The role of injury in SEB development

Results of this study demonstrate that injury enhances the ability of the ciliate, *H. corallasia*, to colonise corals. High proportions (up to 100%) of artificially created injuries were colonised within 10 days of injury in two of the three experimental coral species, highlighting both the rapid colonisation of injuries by ciliates in the field and the ubiquitous presence of ciliates in the early fouling community of bare coral skeleton. The rapid colonisation of injuries by *H. corallasia* (in as little as 6 hours, personal observation) is consistent with the 1–8 h morphogenesis times (time to metamorphose from motile swarmers to sessile trophonts) known for other folliculinid ciliates (Andrews 1914, 1923). Based on observations that *H. corallasia* trophonts contain zooxanthellae engulfed using ciliary currents (Fig. 1f), it is possible that motile swarmers are attracted to sites of recent injury by the release of zooxanthellae from damaged coral tissues.

Despite initial increases in ciliate densities on artificially-inflicted wounds, ciliates remained in loose, scattered

distributions and did not form band-like aggregations associated with tissue lysis on any of the experimental corals. On the contrary, tissues adjacent to injuries regenerated, decreasing the size of lesions through time, suggesting that *H. corallasia* may not be a primary agent of tissue necrosis. If this is the case, then tissue loss recorded on 95% of in situ *A. muricata* colonies with band-like ciliate aggregations on a nearby reef suggests that additional microbial or environmental factors, absent from experimental wounds, may be required before *H. corallasia* aggregations become associated with progressing disease bands. Typically, wildlife diseases involve a web of causation including multiple micro-organisms, nutrition and environmental stressors and very few causative agents are both necessary and sufficient to produce disease over a wide range of conditions (Wobeser 2006). Thus, coral health may need to be compromised in some way before progressing band-like aggregations develop. For example, reduced energetic reserves in bleached or otherwise compromised corals (Grottoli et al. 2004) may limit investment in boundary maintenance or impair wound healing (Meesters and Bak 1993; Fine et al. 2002), increasing the susceptibility of coral hosts to ciliate infections. Shifts in microbial community composition in the mucus of compromised corals (e.g., in bleached corals, Ritchie 2006) may also allow microbes to invade normally resistant coral tissues. Although vectors were unnecessary for colonisation of injuries by *H. corallasia*, corallivores, in particular, may play a role in SEB development through means other than injury creation, for example through the introduction of new microbes during feeding. Experiments on corals with compromised health and involving injuries created by corallivores may help to elucidate factors leading to the development of band-like, potentially pathogenic ciliate aggregations. Microbial studies are also required to evaluate the possibility that a primary causative agent initiates tissue necrosis before *H. corallasia* invades secondarily.

Clear differences in susceptibility to SEB among the three experimental coral species are likely to relate to both ciliate substrate preferences and host resistance mechanisms. Ciliate substrate preferences may reflect differences in the rugosity or density of exposed coral skeleton or differences in colony morphology. The porosity and fine-scale complexity of *P. cylindrica*'s skeletal micro-architecture may provide appropriate interstices within which the ciliates can embed, contributing to the high densities of ciliates found on this species. The more open corallite structure of *P. damicornis* and particularly its high skeletal density (Marshall 2000) may discourage or inhibit ciliate colonisation, contributing to the low densities found on injuries of this species. In addition, greater numbers of *H. corallasia* swarmers may be ingested by coral polyps in the closely intertwining branches of *P. damicornis*, which provide only small gaps through which ciliates can approach injured areas. Conversely, the more open branching patterns of *A. muricata* and *P. cylindrica* may allow *H. corallasia* swarmers to more readily evade polyps and thereby gain easier access to wounds.

Interestingly, differences in susceptibility to SEB among the three coral species did not reflect their capacity for injury repair. In particular, injury regeneration in the presence of ciliates was slowest in *P. cylindrica*, which is a member of the least susceptible family (of the three experimentally injured families) as determined empirically in surveys. Moreover, although ciliate densities were lower on the faster regenerating *P. damicornis*, in general, changes in ciliate densities were not matched by changes in rates of injury regeneration. For example, ciliate densities on *P. damicornis* injuries increased 40 fold between days 10 and 15, over which time injury size decreased by ~70%. Also, ciliate densities on *P. cylindrica* injuries doubled between days 15 and 20, whereas injuries did not change in size. Differences in rates of injury repair among species reflect differing prioritization of energy allocation to maintenance and repair within the life-history strategies of these three species (Hall 1997), but clearly, this is not the main factor driving susceptibility to SEB.

Rather than the capacity for injury repair, vulnerability to injury may be more important in determining differing susceptibilities to SEB among the three coral species. Pocilloporid and acroporid corals, which are the most susceptible coral families on the GBR (Willis et al. 2004), are also amongst the preferred prey of both the crown-of-thorns starfish (*A. planci)* (De'ath and Moran 1998; Pratchett 2007) and *Drupella* snails (Cummings 1999) in the Indo-Pacific. Moreover, fragmentation is a common part of the life history of branching acroporid corals (Wallace 1985). The potential role of injury in disease initiation has been highlighted previously (Antonius 1981, 1985; Bak and Criens 1981) and is likely to underlie correlations between corallivores and disease prevalence in the Red Sea (Antonius and Riegl 1997, 1998). The high susceptibility of pocilloporid corals to SEB in the Red Sea supports the injury vulnerability hypothesis, however acroporid corals were less susceptible than some genera (i.e., *Hydnophora* and *Galaxea*) (Winkler et al. 2004) that are not typically amongst the preferred prey of crown-of-thorns starfish or *Drupella* snails. More detailed analyses of regional differences in taxa susceptibilities to both SEB and injury are needed to further explore the role of injury vulnerability in the establishment of this disease.

The number and impacts of coral diseases have been increasing worldwide in recent years (Harvell et al. 1999; McClanahan et al. 2004; Willis et al. 2004; Miller et al. 2006) and are likely to continue to increase given that there are few options for mitigating their impacts (Bruckner

2002). The identification of injury as a factor that facilitates the colonisation of corals by the ciliate, *H. corallasia*, suggests that protection of reefs from activities which injure corals represents an important management tool. Correlations between increased tourist activity (Winkler et al. 2004) and increased wave action at shallow sites (Antonius and Lipscomb 2001) with increased prevalence of SEB provide corroborative evidence of the importance of reef protection as a management strategy. In particular, management of activities which result in coral breakage (e.g., destructive fishing practices, snorkelling, diving, reef walking, and boat groundings) is an important strategy for minimising the transmission and spread of SEB on local scales.

In summary, SEB is endemic within the GBR, affecting ~2% of corals in recent surveys, and currently has the widest host range documented for any coral disease. Reported here are measured rates of SEB progression (average 2 ± 0.3 [± SE] mm/day), which, although low in comparison to rates of progression of other GBR diseases, are nevertheless equivalent to or faster than progression rates of diseases that have had significant impacts on Caribbean reefs. Injury facilitates the colonisation of otherwise healthy corals by *H. corallasia*, the putative pathogen of SEB. Vectors were not required for ciliate colonisation of injuries in aquaria, suggesting that their motile phases disperse in the water column. However, the lack of development of progressing, band-like aggregations of ciliates on experimental injuries in the field suggests that either *H. corallasia* is not the causative agent of tissue loss, or, additional factors that increase its virulence or reduce host resistance are required before ciliate infections become pathogenic. Microbial studies are needed to elucidate the potential role of an alternative pathogen in tissue lysis associated with band-like aggregations of ciliates.

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References

Andrews EA (1914) Distribution of folliculina in 1914. Biol Bull 29:373–380

Andrews EA (1923) Folliculina: case making, anatomy and transformation. J Morphol 38:207–278

Antonius A (1981) Coral reef pathobiology: a review. Proc 4th Int Coral Reef Symp 2:3–6

Antonius A (1985) Black band disease infection experiments on hexacorals and octocorals. Proc 5th Int Coral Reef Congr 6:155–160

Antonius A (1999) *Halofolliculina corallasia*, a new coral killing ciliate on Indo-Pacific reefs. Coral Reefs 18:300

Antonius A, Riegl B (1997) A possible link between coral disease and a corallivorous snail (*Drupella cornus)* outbreak in the Red Sea. Atoll Res Bull 47:1–9

Antonius A, Riegl B (1998) Coral diseases and *Drupella cornus* invasion in the Red Sea. Coral Reefs 17:48

Antonius A, Lipscomb D (2001) First protozoan coral-killer identified in the Indo-Pacific. Atoll Res Bull 481–493:1–21

Bak RPM, Criens SR (1981) Survival after fragmentation of colonies of *Madacis mirabilis, Acropora palmata* and *A.cervicornis* (Scleractinia) and the subsequent impact of a coral disease. Proc 4th Int Coral Reef Symp 2:221–227

Boyett H, Bourne D, Willis BL (2007) Elevated temperatures and light enhance progression and spread of black band disease on staghorn corals of the Great Barrier Reef. Mar Biol 151:1711–1720

Bruckner AW (2002) Priorities for effective management of coral diseases. NOAA Technical Memorandum. NMFS-OPR-22. NOAA National Marine Fisheries Service, Silver Springs

Cervino J, Goreau T, Nagelkerken I, Smith GW, Hayes R (2001) Yellow band and dark spot syndromes in Caribbean coral: distribution, rate of spread, cytology, and the effects on abundance and division rate of zooxanthellae. Hydrobiologia 460:53–63

Cróquer A, Bastidas C, Lipscomb D (2006a) Folliculinid ciliates: a new threat to Caribbean corals? Dis Aquat Org 69:75–78

Cróquer A, Bastidas C, Lipscomp D, Rodríguez-Martínez RE, Jordan-Dahlgren E, Guzman HM (2006b) First report of folliculinid ciliates affecting Caribbean scleractinian corals. Coral Reefs 25:187–191

Cummings RL (1999) Predation on reef-building corals: multiscale variation in the density of three corallivorous gastropods, *Drupella* spp. Coral Reefs 18:147–157

Dalton SJ, Smith SDA (2006) Coral disease dynamics at a subtropic location, Solitary Islands Marine Park, Eastern Australia. Coral Reefs 25:37–45

De'ath G, Moran PJ (1998) Factors affecting the behaviour of crown-of-thorns starfish (*Acanthaster planci* L.) on the Great Barrier Reef. 2 Feeding preferences. J Exp Mar Biol Ecol 220:107–126

Fine M, Oren U, Loya Y (2002) Bleaching effect on regeneration and resource translocation in the coral *Oculina patagonica*. Mar Ecol Prog Ser 234:119–125

Garzon-Ferreira J, Zea S (1992) A mass mortality of *Gorgonia ventalina* (Cnidaria: Gorgoniidae) in the Santa Marta area, Caribbean coast of Columbia. Bull Mar Sci 50:522–526

Green E, Bruckner A (2000) The significance of coral disease epizootiology for coral reef conservation. Biol Conserv 96:347–361

Grottoli AG, Rodrigues LJ, Juarez C (2004) Lipids and stable carbon isotopes in two species of Hawaiian corals, *Porites compressa* and *Montipora verrusosa*, following a bleaching event. Mar Biol 145:621–631

Guzman HM, Cortes J (1984) Mortandad de *Gorgonia flabellum* Linnaeus (Octocorallia:Gorgoniidae) en la costa Caribe de Costa Rica. Rev Biol Trop 32:305–308

Hall V (1997) Effects of injury on growth, reproduction and survivorship for common reef-crest corals. Proc 8th Int Coral Reef Symp 1:571–574

Harvell CD, Jordan-Dahlgren E, Merkel SM, Rosenberg E, Raymundo LJ, Smith G, Weil E, Willis BL (2007) Coral disease, environmental drivers and the balance between coral and microbial associates. Oceanography 20:36–59

Harvell CD, Kim K, Burkholder J, Colwell RR, Epstein PR, Grimes J, Hofmann EE, Lipp EK, Osterhaus ADME, Overstreet R, Porter JW, Smith GW, Vasta GR (1999) Emerging marine diseases—climate links and anthropogenic factors. Science 285:1505–1510

Kim K, Harvell CD (2004) The rise and fall of a six year coral-fungal epizootic. Am Nat 164:S52–S63

Loya Y (2004) Coral reefs of Eliat: past, present and future. In: Rosenberg E, Loya Y (eds) Coral disease and health. Springer, Berlin, pp 1–34

Marshall PA (2000) Skeletal damage in reef corals: relating resistance to colony morphology. Mar Ecol Prog Ser 200:177–189

McClanahan TR, McLaughlin SM, Davy JE, Wilson WH, Peters EC, Price KL, Maina J (2004) Observations of a new source of coral mortality along the Kenyan coast. Hydrobiologia 530:469–479

McCullagh P, Nelder JA (1989) Generalized linear models. Chapman Hall, London

Meesters EH, Bak RPM (1993) Effects of coral bleaching on tissue regeneration potential and coral survival. Mar Ecol Prog Ser 96:189–198

Miller J, Waara R, Muller E, Rogers C (2006) Coral bleaching and disease combine to cause extensive mortality on reefs in US Virgin Islands. Coral Reefs 25:418

Nagelkerken I, Buchan K, Smith G, Boniar K, Bush P, Garzon-Ferreira J, Botero L, Gayle P, Heberer C, Petrovic C, Pors L, Yoshioka P (1997) Widespread disease in Caribbean Sea Fans: I Spreading and general characteristics. Proc 8th Int Coral Reef Symp 1:679–682

Page CA, Willis BL (2006) Distribution, host range and large-scale variability in the prevalence of black band disease on the Great Barrier Reef, Australia. Dis Aquat Org 69:41–51

Patterson KL, Porter JW, Ritchie KB, Polson SW, Mueller E, Peters EC, Santavy DL, Smith GW (2002) The etiology of white pox, a lethal disease of the Caribbean elkhorn coral, *Acropora palmata*. Proc Natl Acad Sci USA 99:8725–8730

Porter J, Dustan P, Jaap W, Patterson KL, Kosmynin V, Meier O, Patterson M, Parsons M (2001) Patterns of spread of coral disease in the Florida Keys. Hydrobiologia 460:1–24

Pratchett MS (2007) Feeding preferences of *Acanthaster planci* (Echinodermata: Asteroidea) under controlled conditions of food availability. Pac Sci 61:113–120

Precht WF, Bruckner AW, Aronson RB, Bruckner RJ (2002) Endangered acroporid corals of the Caribbean. Coral Reefs 21:41–42

Raymundo LJ, Rosell KB, Reboton CT, Kaczmarsky L (2005) Coral disease on Philippine reefs: genus *Porites* is a dominant host. Dis Aquat Org 64:181–191

Raymundo LJH, Harvell CD, Reynolds TL (2003) *Porites* ulcerative white spot disease: description, prevalence, and host range of a new coral disease affecting Indo-Pacific reefs. Dis Aquat Org 56:95–104

Riegl B, Antonius A (2003) Halofolliculina skeleton eroding band (SEB): a coral disease with fossilization potential? Coral Reefs 22:48

Ritchie KB (2006) Regulation of microbial populations by coral surface mucus and mucus-associated bacteria. Mar Ecol Prog Ser 322:1–14

Roff G, Hoegh-Guldberg O, Fine M (2006) Intra-colonial response to Acroporid "white syndrome" lesions in tabular *Acropora* spp. (Scleractinia). Coral Reefs 25:255–264

Sutherland KP, Porter JW, Torres C (2004) Disease and immunity in Caribbean and Indo-Pacific zooxanthellate corals. Mar Ecol Prog Ser 266:273–302

Wallace CC (1985) Reproduction, recruitment and fragmentation in nine sympatric species of the coral *Acropora*. Mar Biol 88:217–233

Ward JR, Lafferty KD (2004) The elusive baseline of marine disease: are diseases in ocean ecosystems increasing? PLoS Biology 2:542–547

Weil E (2004) Coral reef disease in the wider Caribbean: status and prognosis. In: Rosenberg E, Loya Y (eds) Coral disease and health. Springer, Berlin, pp 35–64

Weil E, Smith G, Gil-Agudelo DL (2006) Status and progress in coral reef disease research. Dis Aquat Org 69:1–7

Williams Jr EH, Bartels PJ, Bunkley-Williams L (1999) Predicted disappearance of coral-reef ramparts: a direct result of major ecological disturbances. Global Change Biol 5:839–845

Willis BL, Page CA, Dinsdale EA (2004) Coral disease on the Great Barrier Reef. In: Rosenberg E, Loya Y (eds) Coral disease and health. Springer, Berlin, pp 69–104

Winkler R, Antonius A, Abigail Renegar D (2004) The skeleton eroding band disease on coral reefs of Aqaba, Red Sea. PSZNI Mar Ecol 25:129–144

Wobeser GA (2006) Essentials of disease in wild animals. Blackwell, Ames

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